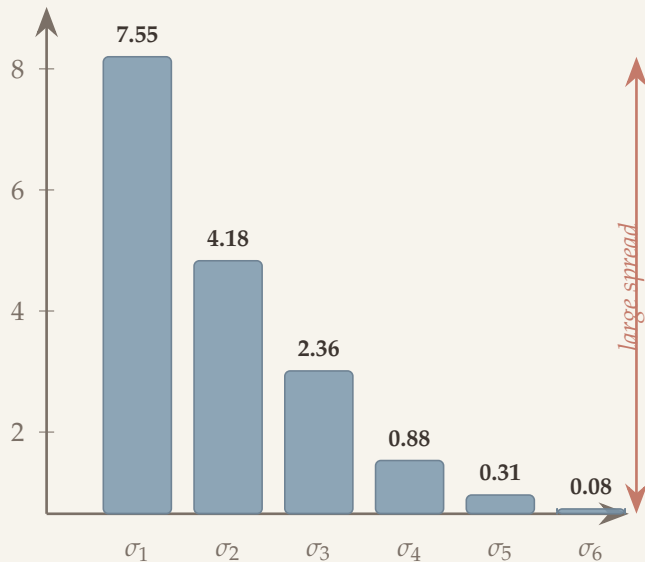
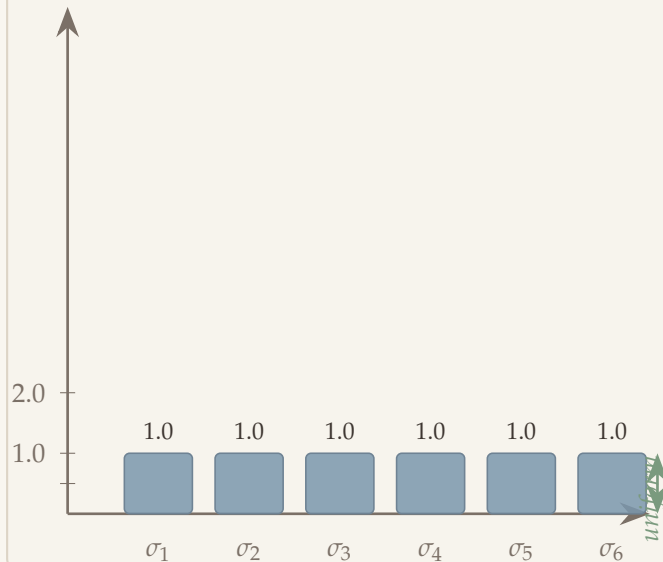


Gradient G: singular values



Muon update $UV^\top$: singular values



Muon normalizes all singular values to 1, ensuring uniform step scale across all directions